LEAP 2025

Mathematics

Practice Test

Grade 8

DEPARTMENT of EDUCATION
Louisiana Believes

Session 1

Directions:

Today, you will take Session 1 of the Grade 8 Mathematics Practice Test. You will not be able to use a calculator in this session.

Read each question. Then, follow the directions to answer each question. Mark your answers by circling the correct choice. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the space provided in your test booklet.

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this session **ONLY**.

GO ON ►

1. Which of these expressions represent solutions to the equation $y^3 = 64$?

Select **each** correct answer.

A. $^{-}\sqrt[3]{64}$

B. $\sqrt[3]{64}$

C. $^{-}8$

D. 8

E. $^{-}4$

F. 4

GO ON ▶

2. Which decimal is the equivalent of $\frac{6}{11}$?

A. $0.18\overline{3}$

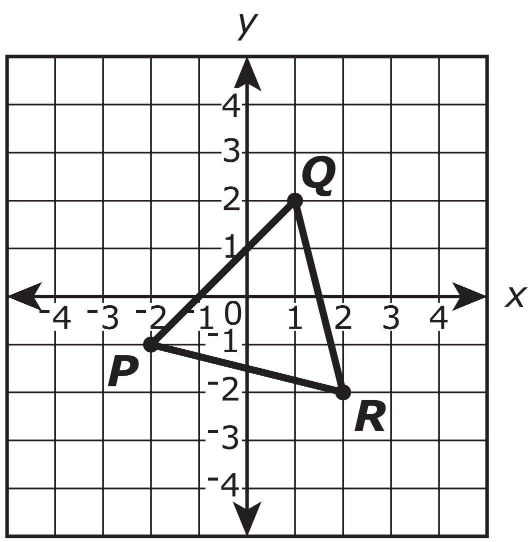
B. $0.1\overline{83}$

C. $0.5\overline{4}$

D. $0.\overline{54}$

GO ON ▶

3. Triangle *PQR* is shown on the coordinate plane.



Triangle *PQR* is rotated 90° counterclockwise about the origin to form the image triangle *P′Q′R′* (not shown). Then triangle *P′Q′R′* is reflected across the *x*-axis to form triangle *P″Q″R″* (not shown).

Part A

What are the signs of the coordinates (x, y) of point *P′*?

A. Both x and y are positive.

B. x is negative and y is positive.

C. Both x and y are negative.

D. x is positive and y is negative.

Part B

What are the signs of the coordinates (x, y) of point *Q″*?

A. Both x and y are positive.

B. x is negative and y is positive.

C. Both x and y are negative.

D. x is positive and y is negative.

GO ON ►

4. Which of the input-output tables represent a function?

Select **each** correct answer.

A.

Input	**Output**
1	4
1	6
5	5
8	10

B.

Input	**Output**
1	4
5	6
5	1
10	8

C.

Input	**Output**
1	4
8	6
5	1
10	5

D.

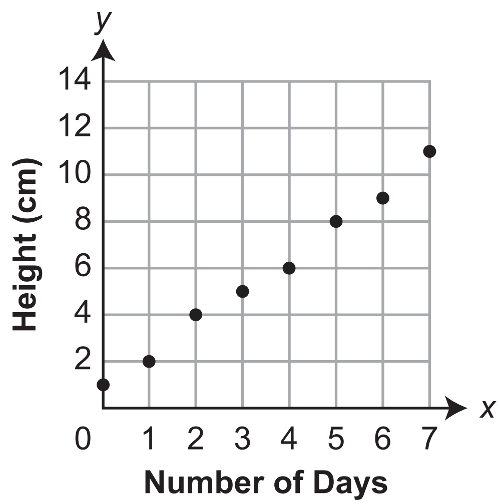
Input	**Output**
1	4
10	6
5	5
8	1

E.

Input	**Output**
1	4
8	6
5	10
1	5

GO ON ►

5. Points are shown plotted on the coordinate plane. The points represent a relation, where x is the input and y is the output.



Complete the sentence to explain whether or not this set of points represents a function.

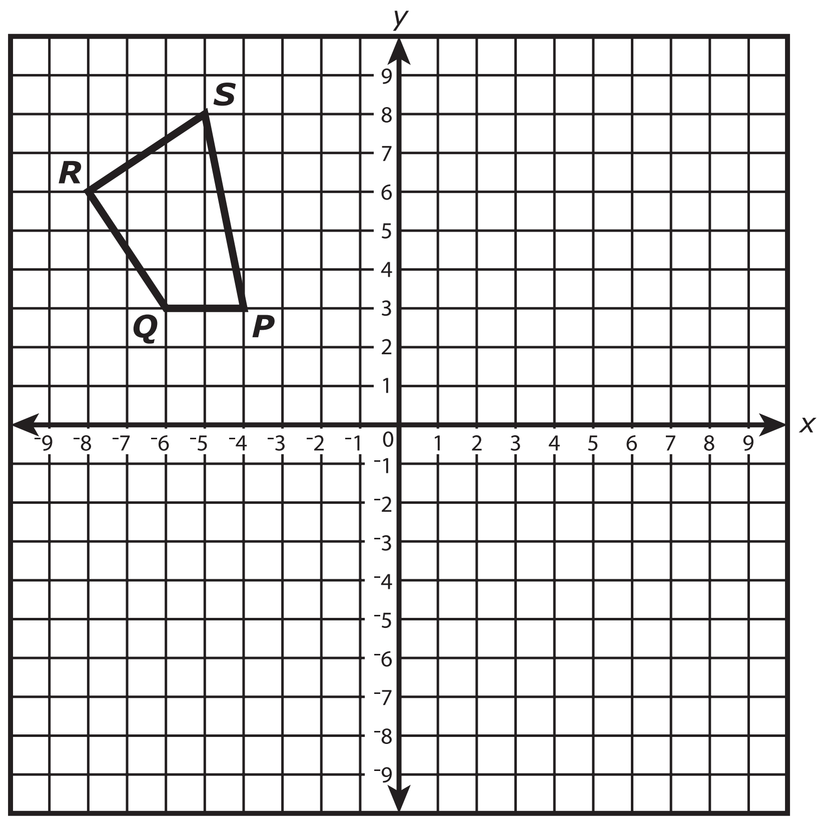
Select the appropriate phrase from each list to complete the sentence.

It ______________________________ because ______________________________ .

does represent a function	each input has only one output
does not represent a function	each output has only one input
	one input has two outputs
	one output has two inputs

GO ON ▶

6. Polygon *KLMN* is the image of polygon *PQRS* after a 180° rotation.



Which angle of polygon *KLMN* is congruent to $\angle S$?

A. $\angle K$

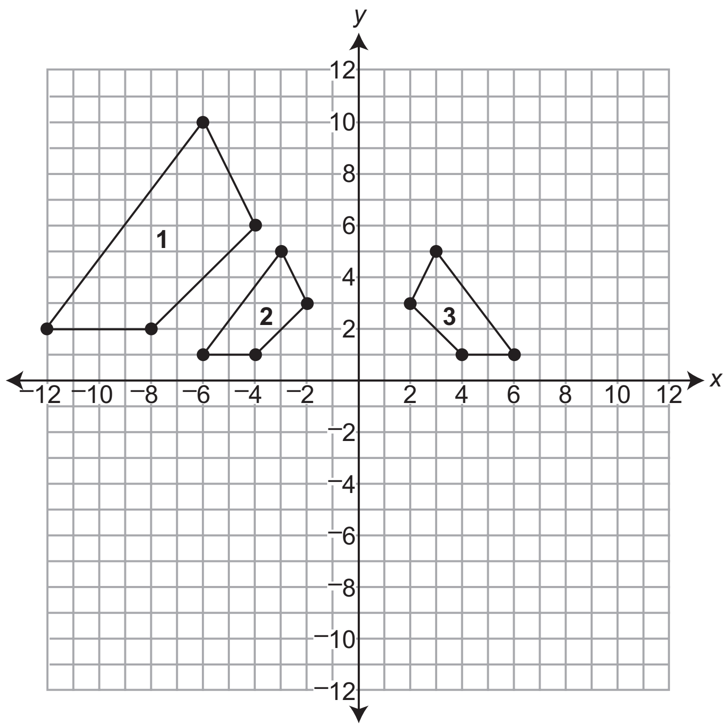
B. $\angle L$

C. $\angle M$

D. $\angle N$

GO ON ►

7. On the coordinate plane shown, Figure 1 is transformed into Figure 2, which is transformed into Figure 3. Figure 1 and Figure 3 are similar by a sequence of transformations.



Part A

What type of transformation was used to transform Figure 1 into Figure 2?

A. dilation

B. reflection

C. rotation

D. translation

GO ON ►

Part B

Which statement describes the transformation of Figure 2 into Figure 3?

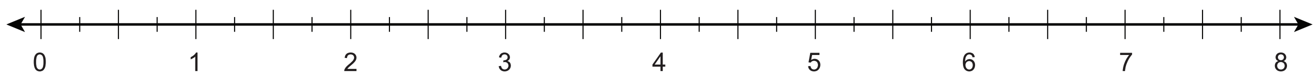
A. reflection across the *x*-axis

B. reflection across the *y*-axis

C. translation 4 units to the right

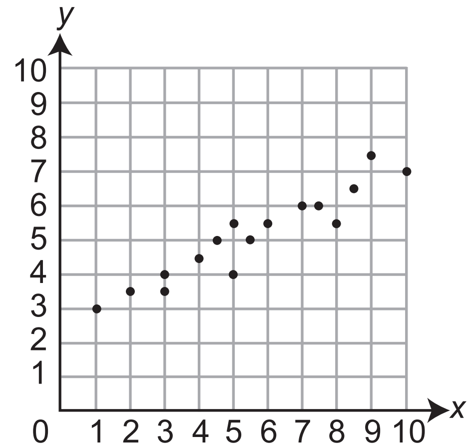
D. translation 6 units to the right

8. Plot a point on the number line that **best** approximates the location of $\sqrt{14}$.



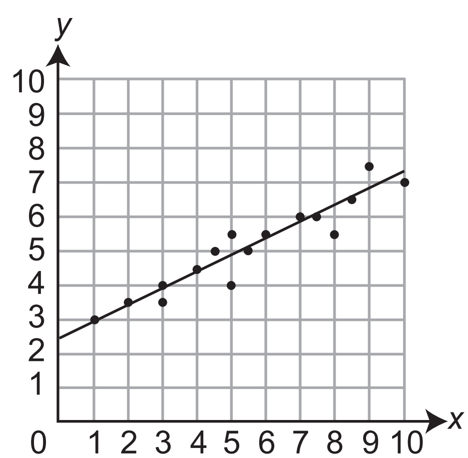
GO ON ▶

9. A scatter plot is shown on the coordinate plane.

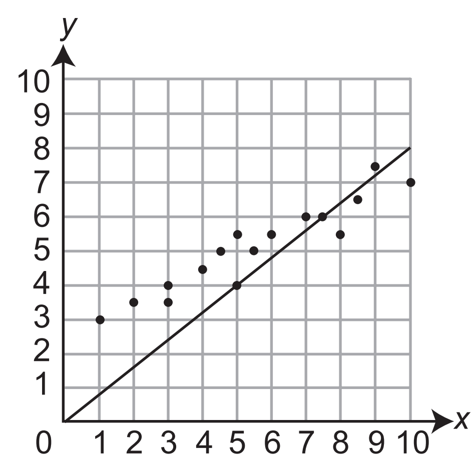


Which of these **most closely** approximates a line of best fit for the data in the scatter plot?

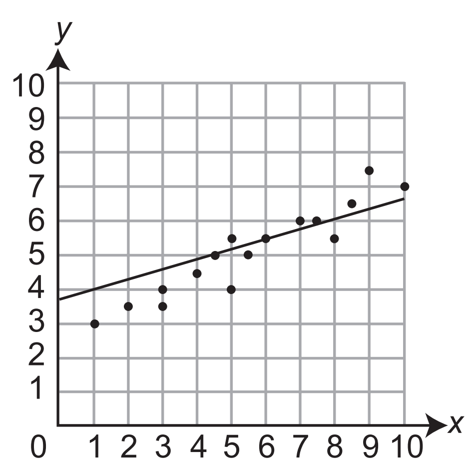
A.



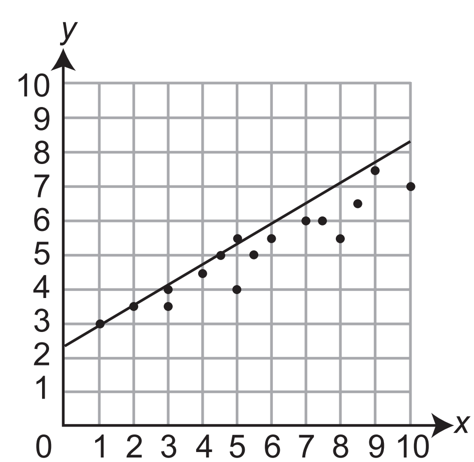
B.



C.



D.



GO ON ▶

10. The body of a 154-pound person contains approximately 2×10^{-1} milligrams of gold and 6×10^{1} milligrams of aluminum. Based on this information, the number of milligrams of aluminum in the body is how many times the number of milligrams of gold in the body?

Enter your answer in the box.

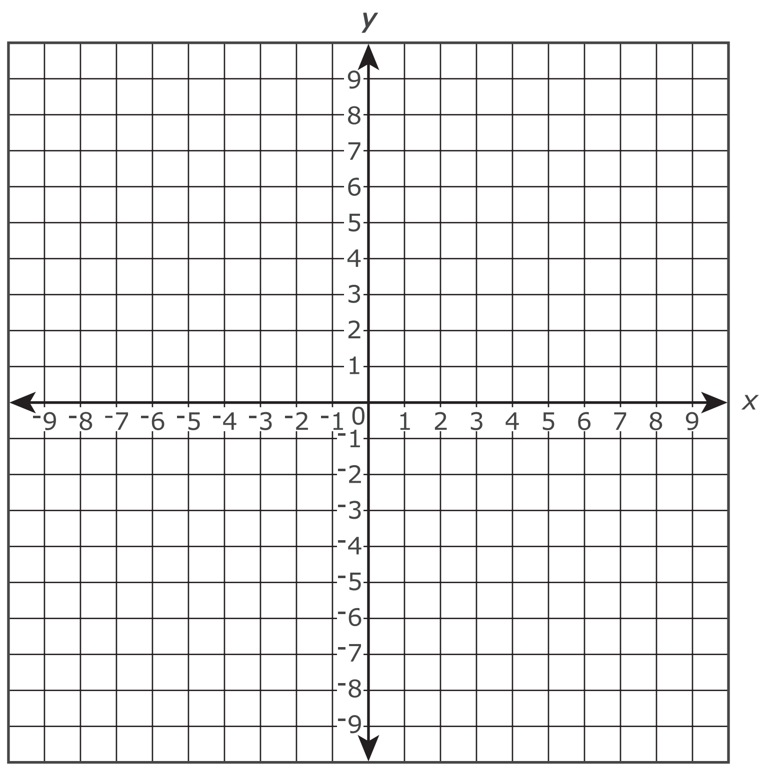
GO ON ▶

11. The equation of line *s* is $y = \frac{1}{3}x - 3$.

The equation of line *t* is $y = {}^{-}x + 5$.

The equations of lines *s* and *t* form a system of equations. The solution to the system of equations is located at point P.

Draw a line to represent line *s* and another line to represent line *t*. Then plot point P.



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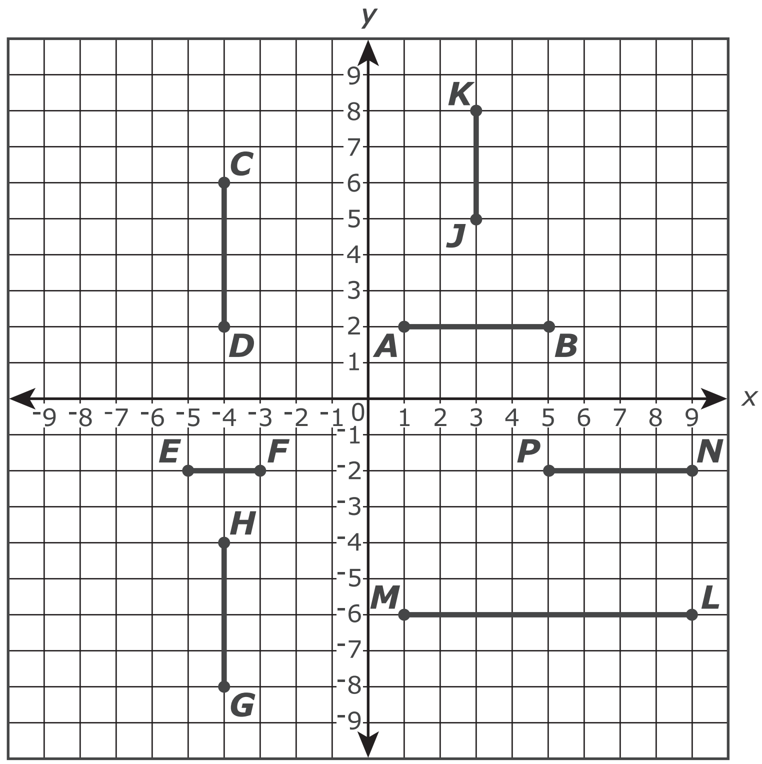


12. Seven line segments are shown on the coordinate plane.

Which of these segments could be the image of segment *AB* after a sequence of reflections, rotations, and/or translations?

Select **each** correct answer.

A. line segment *CD*

B. line segment *EF*

C. line segment *GH*

D. line segment *JK*

E. line segment *LM*

F. line segment *NP*

GO ON ►

13. What value of x makes the equation $3(x - 6) - 8x = {}^{-}2 + 5(2x + 1)$ true?

Enter your answer in the box.

GO ON ▶

14. A system of equations is shown.

$$\begin{cases} x = 10 \\ 3x + 5y = 20 \end{cases}$$

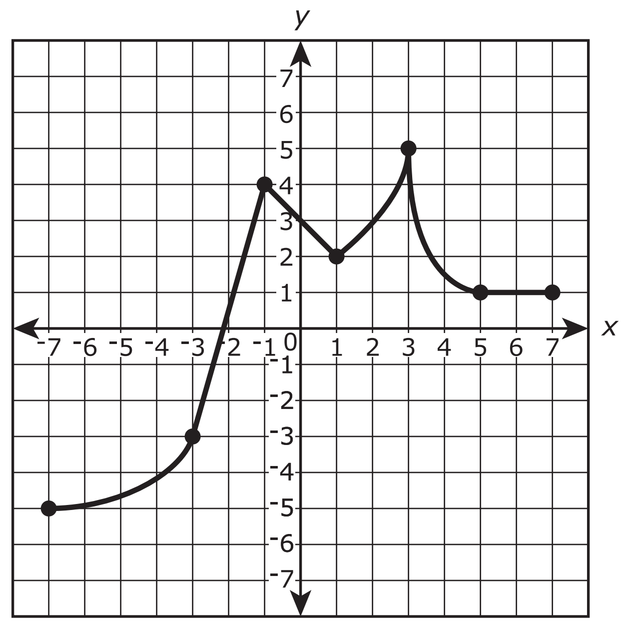
What is the solution (x, y) of the system of equations?

Enter your answers in the boxes.

(☐ , ☐)

GO ON ▶

15. The graph shows *y* as a function of *x*.



For each interval in the table, mark an X in the appropriate box to indicate whether the function is increasing, decreasing, or neither increasing nor decreasing over the interval.

	Increasing	**Decreasing**	**Neither Increasing nor Decreasing**
$^{-}7 < x < {}^{-}3$			
$^{-}3 < x < {}^{-}1$			
$^{-}1 < x < 1$			
$1 < x < 3$			
$3 < x < 5$			
$5 < x < 7$			

GO ON ►

16. A system of two linear equations is graphed on a coordinate plane. If the system of equations has infinitely many solutions, which statement must be true?

A. On the graph, there are no points (x, y) that satisfy both equations.

B. On the graph, there is exactly one point (x, y) that satisfies both the equations.

C. On the graph, any point (x, y) that satisfies one of the equations cannot satisfy the other equation.

D. On the graph, any point (x, y) that satisfies one of the equations must also satisfy the other equation.

GO ON ▶

17. Which expression is equivalent to $\frac{2^{-3}}{2^{-5}}$?

A. 2^2

B. $\frac{1}{2^2}$

C. 2^8

D. $\frac{1}{2^8}$

GO ON ▶

18. Mark an X in the appropriate box to classify each equation as defining *y* as a linear or nonlinear function of *x*. Select one cell per column.

	$y = 7 \times 4x$	$y = (2x + 5)^2$	$y = 10x^2$	$y = 5x - 3$	$y = \frac{x}{2}$	$y = 2x^3 + 1$
linear						
nonlinear						

GO ON ►

19. A carpenter bought 750 nails. Each nail has a mass of 5.2×10^{-3} kilogram. What is the total mass, in kilograms, of the nails the carpenter bought? Give your answer as a decimal.

Enter your answer in the box.

GO ON ▶

20. $\frac{3}{4}(x + 8) = 9$

In the equation shown, what is the value of *x* that makes the equation true?

Enter your answer in the box.

STOP

Mathematics—Session 2

Session 2

Directions:

Today, you will take Session 2 of the Grade 8 Mathematics Practice Test. You will be able to use a calculator in this session.

Read each question. Then, follow the directions to answer each question. Mark your answers by circling the correct choice. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the space provided in your test booklet.

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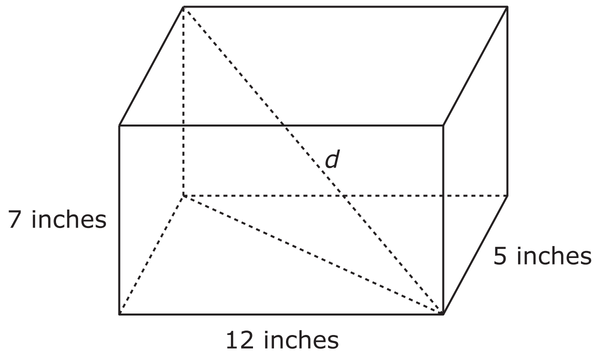
GO ON ►

21. A right rectangular prism is shown.

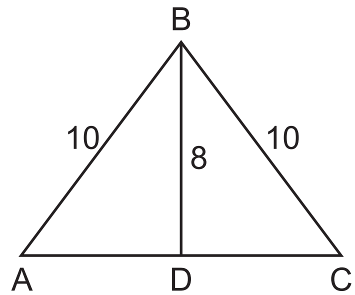


To the nearest thousandth of an inch, what is the length of the diagonal, *d*?

Enter your answer in the box.

GO ON ►

22. In $\triangle ABC$, $\overline{BD}$ is perpendicular to $\overline{AC}$. The dimensions are shown in centimeters.



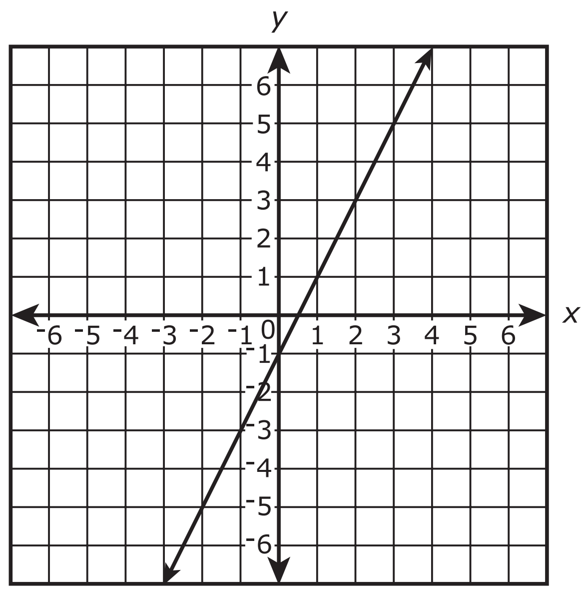
What is the length, in centimeters, of $\overline{AC}$?

Enter your answer in the box.

GO ON ▶

23. The graph of Function 1 is shown on the coordinate plane.

Function 1



The equations of three other functions are given.

Function 2	Function 3	Function 4
$y = 3 + 2x$	$y = 2$	$y = \frac{3}{2}x + 6$

Which function or functions have a slope equal to the slope of Function 1?

A. Function 2 only

B. Function 4 only

C. Function 2 and Function 3 only

D. Function 2 and Function 4 only

GO ON ▶

24. A tank of water was drained at a constant rate. The table shows the number of gallons of water left in the tank after being drained for two amounts of time.

Draining Time (minutes)	Water in Tank (gallons)
10	450
30	330

Part A

What is the rate at which the water was drained from the tank?

A. 6 gallons of water per minute

B. 11 gallons of water per minute

C. 45 gallons of water per minute

D. 120 gallons of water per minute

Part B

What was the total amount of water in the tank before it was drained?

A. 450 gallons

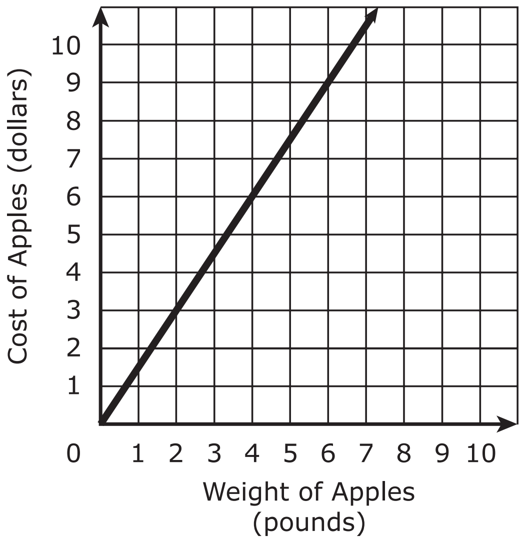
B. 510 gallons

C. 560 gallons

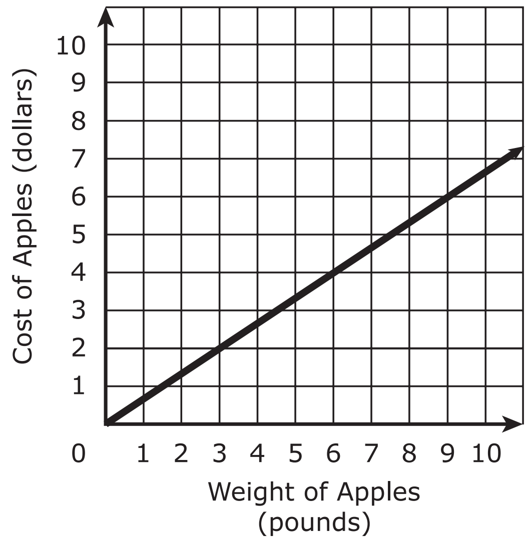
D. 570 gallons

25. At a local market, the cost of apples is directly proportional to the weight of the apples. Carlos bought 10 pounds of apples for a cost of $15.00. Which graph shows the relationship between the weight of the apples, in pounds, and the cost of the apples, in dollars?

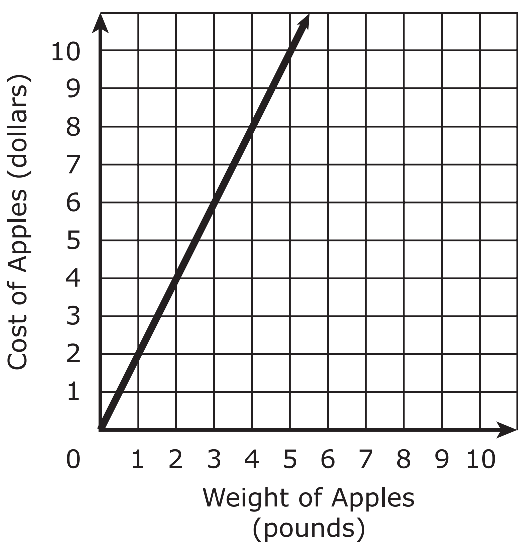
A.



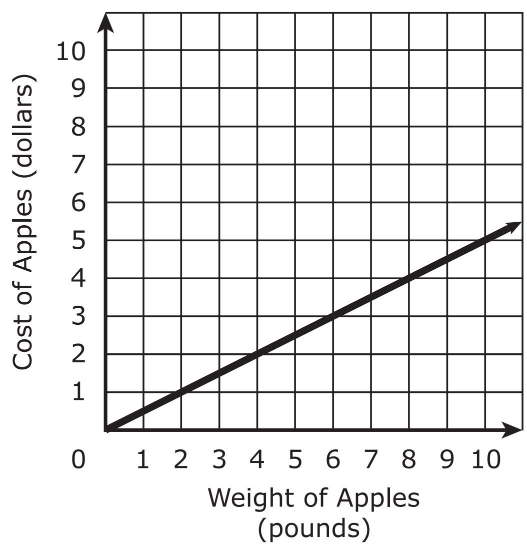
B.



C.



D.



GO ON ▶

26. The table shows the results of a random survey of students in grade 7 and grade 8. Every student surveyed gave a response. Each student was asked if he or she exercised less than 5 hours last week or 5 or more hours last week.

	Less than 5 Hours	**5 or More Hours**
Grade 7 Students	49	63
Grade 8 Students	58	51

Based on the results of the survey, which statements are true?

Select **each** correct statement.

A. More grade 8 students were surveyed than grade 7 students.

B. A total of 221 students were surveyed.

C. Less than 50% of the grade 8 students surveyed exercised 5 or more hours last week.

D. More than 50% of the students surveyed exercised less than 5 hours last week.

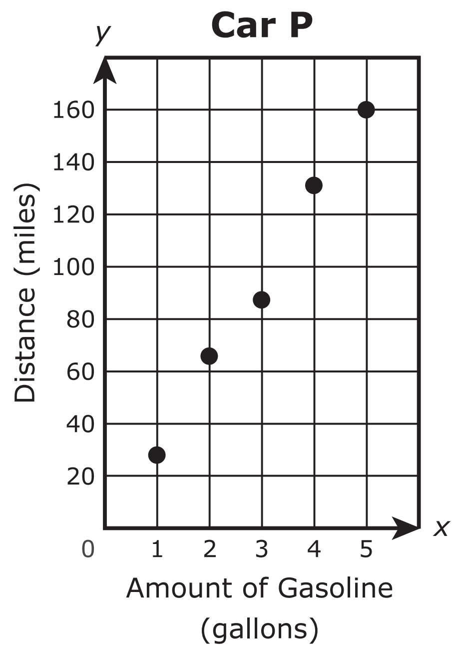
E. A total of 107 grade 7 students were surveyed.

GO ON ►

27. The gasoline mileage for two cars can be compared by finding the distance each car traveled and the amount of gasoline used. The table shows the distance that car M traveled using *x* gallons of gasoline. The graph shows the distance, *y*, that car P traveled using *x* gallons of gasoline.

Car M

Amount of Gasoline (gallons)	Distance (miles)
2	50.4
3	80.5
7	181.3
5	137.5



Based on the information in the table and the graph, compare the approximate miles per gallon of car M to car P. Show your work or explain your answer.

Enter your answer and your work or explanation in the box provided.

GO ON ▶

28. Peter determined the area, in square miles, of a piece of land using his calculator. The result of his calculation is displayed on his calculator in scientific notation as 7.4e-4.

Which statement is true of the area of the piece of land?

A. It is between 0.07 and 0.7 square mile.

B. It is between 0.007 and 0.07 square mile.

C. It is between 0.0007 and 0.007 square mile.

D. It is between 0.00007 and 0.0007 square mile.

GO ON ▶

29. Two different proportional relationships are represented by the equation and the table.

Proportion A

$y = 9x$

Proportion B

x	y
0	0
3	34.5
5	57.5
8	92

Select from each list to complete the sentence comparing the rates of change of the proportional relationships.

The rate of change in Proportion A is __________ __________ than the rate of change in Proportion B.

1.5	more
2.5	less
25.5	
43.5	

GO ON ▶

30. Determine whether the equation has no solution, one solution, or infinitely many solutions.

$^{-}2(11 - 12x) = {}^{-}4(1 - 6x)$

Show each step of your work. Explain your conclusion.

Enter your answer, your work, and your explanation in the box provided.

GO ON ▶

31. Martin is considering the expressions $\frac{1}{2}(7x + 48)$ and $-\left(\frac{1}{2}x - 3\right) + 4(x + 5)$. He wants to know if one expression is greater than the other for all values of x.

Part A

Which statement about the relationship between the expressions is true?

A. The value of the expression $\frac{1}{2}(7x + 48)$ is always equal to the value of the expression $-\left(\frac{1}{2}x - 3\right) + 4(x + 5)$.

B. The value of the expression $\frac{1}{2}(7x + 48)$ is always less than the value of the expression $-\left(\frac{1}{2}x - 3\right) + 4(x + 5)$.

C. The value of the expression $\frac{1}{2}(7x + 48)$ is always greater than the value of the expression $-\left(\frac{1}{2}x - 3\right) + 4(x + 5)$.

D. The value of the expression $\frac{1}{2}(7x + 48)$ is sometimes greater than and sometimes less than the value of the expression $-\left(\frac{1}{2}x - 3\right) + 4(x + 5)$.

GO ON ►

Part B

Show or explain how you found your answer to Part A.

Enter your work or your explanation in the box provided.

Part C

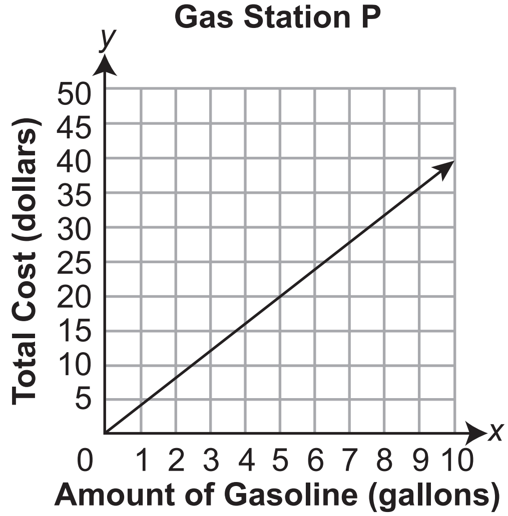
Write a new expression that always has a greater value than both of these expressions.

Enter your expression in the box provided.

GO ON ▶

32. The graph and table show the amount of gasoline in gallons, x, and total cost in dollars, y, of gasoline at two gas stations.

Gas Station P



Gas Station M

x	y
5	19.00
10	38.00
15	57.00

Use the unit price of gasoline at both gas stations to determine which gas station charges more for gasoline (gallons). Be sure to include the unit prices in your answer. Show or explain your work.

Enter your answer and your work or explanation in the box provided.

STOP

Session 3

Directions:

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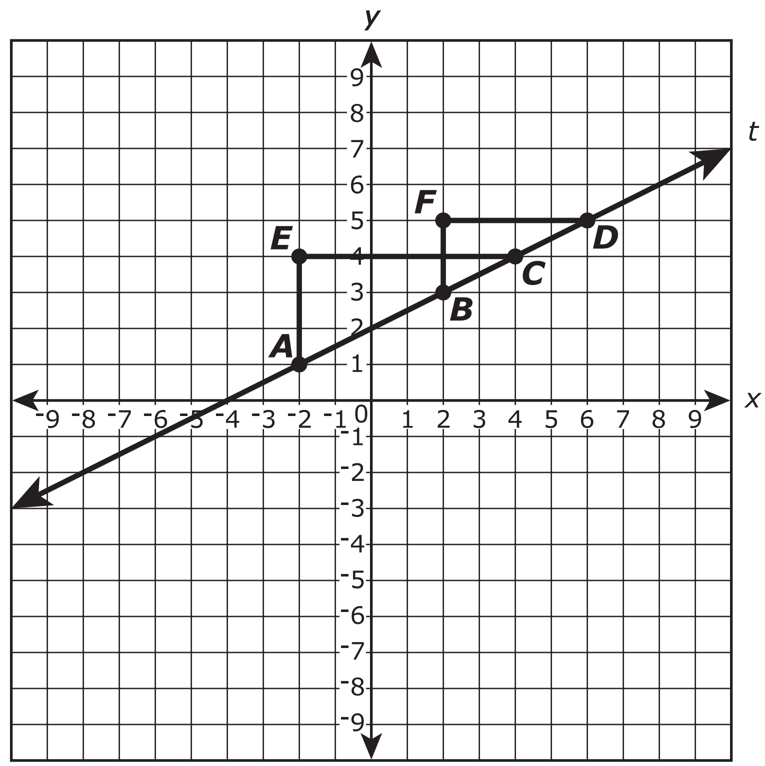
If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this session **ONLY**.

GO ON ▶

33. Line t and $\triangle ECA$ and $\triangle FDB$ are shown on the coordinate plane.



GO ON ►

Which statements are true?

Select **all** that apply.

A. The slope of $\overline{AC}$ is equal to the slope of $\overline{BC}$.

B. The slope of $\overline{AC}$ is equal to the slope of $\overline{BD}$.

C. The slope of $\overline{AC}$ is equal to the slope of line t.

D. The slope of line t is equal to $\frac{EC}{AE}$.

E. The slope of line t is equal to $\frac{FB}{FD}$.

F. The slope of line t is equal to $\frac{AE}{FD}$.

GO ON ▶

34. A survey of 7th and 8th grade students asked whether they were in favor of or against school uniforms. The two-way table shows the results.

Survey Results

Grade	Number of Students		
	In Favor	**Against**	**Total**
7th	48	64	112
8th	68	70	138
Total	116	134	250

To the nearest tenth of a percent, what percent of the 7th grade students were in favor of wearing school uniforms?

A. 19.2%

B. 41.3%

C. 42.9%

D. 57.1%

GO ON ►

35. Filipo is building a rectangular sandbox for his younger brother. The length of the sandbox is 1 foot longer than twice the width of the sandbox. The perimeter of the sandbox is 29 feet.

Part A

Which equation could be used to determine w, the width, in feet, of the sandbox?

A. $w + w + 2 = 29$

B. $w + 2w + 1 = 29$

C. $2w + 2(w + 2) = 29$

D. $2w + 2(2w + 1) = 29$

Part B

What is the width, in feet, of the sandbox?

Enter your answer in the box.

GO ON ▶

36. Functions A, B, and C are linear functions.

Some values of Function A are shown in the table.

Function A

x	y
3	3
5	7
6	9

The graph of Function B has a y-intercept of (0, 3) and an x-intercept of ($^{-}5$, 0).

Function C is defined by the equation $y = (3x + 1)$.

Order the linear functions based on rate of change, from least to greatest.

Functions:

Function A

Function B

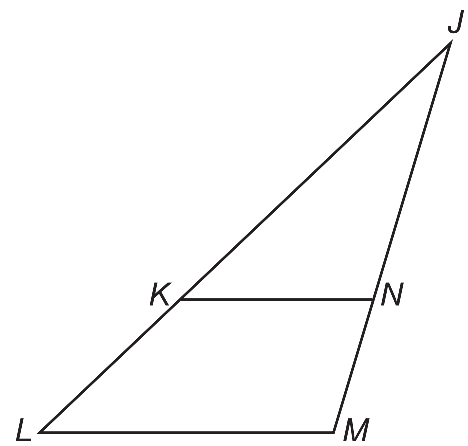
Function C

Least Rate

Greatest Rate

GO ON ▶

37. In the figure shown, $\overline{KN}$ is parallel to $\overline{LM}$.



Part A

When comparing $\triangle KJN$ and $\triangle LJM$, Tara states that $\angle KJN$ and $\angle LJM$ are congruent.

Explain why Tara's statement is correct.

Enter your explanation in the box provided.

GO ON ▶

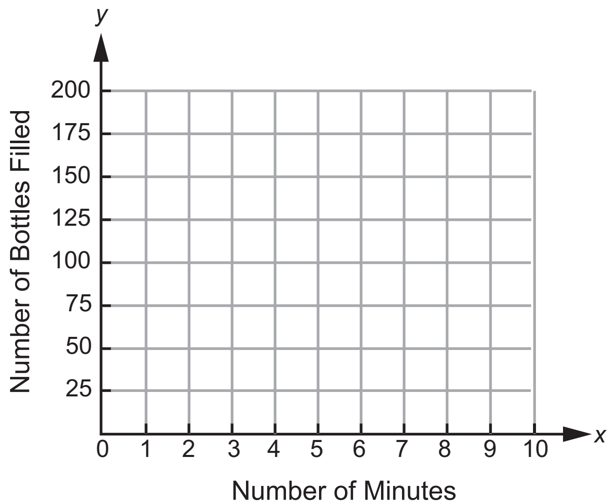
Part B

Tara wants to prove that a second pair of corresponding angles from $\triangle KJN$ and $\triangle LJM$ are congruent. Determine a second pair of corresponding angles from $\triangle KJN$ and $\triangle LJM$ that are congruent. Then explain how you know that the two angles are congruent.

Enter your answer and your explanation in the box provided.

GO ON ►

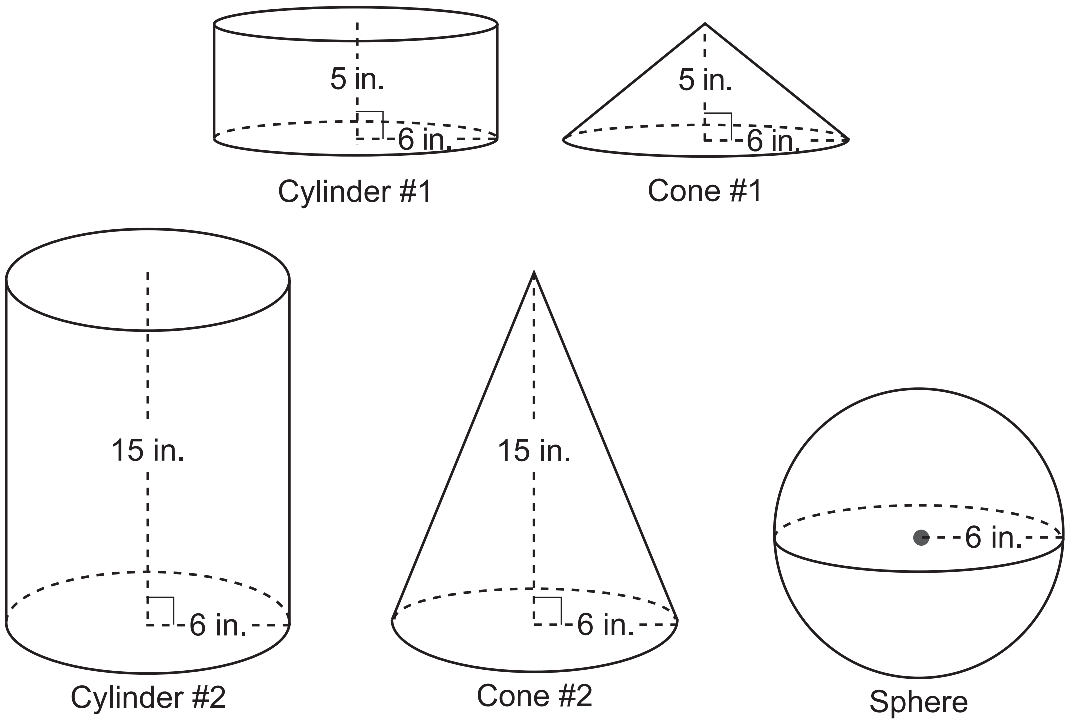
38. The number of bottles a machine fills is proportional to the number of minutes the machine operates. The machine fills 250 bottles every 20 minutes. Create a graph that shows the number of bottles, y, the machine fills in x minutes.



GO ON ▶

Consider the figures shown to answer Part A and Part B.

39.



Part A

Which figures have a volume greater than 600 cubic inches?

Select **all** that apply.

A. Cylinder #1

B. Cone #1

C. Cylinder #2

D. Cone #2

E. Sphere

GO ON ►

Part B

How many times greater is the volume of the Sphere than the volume of Cone #1? Round your answer to the nearest tenth.

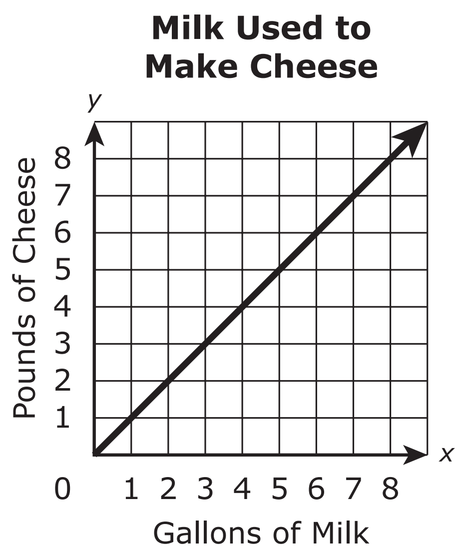
Enter your answer in the box.

GO ON ▶

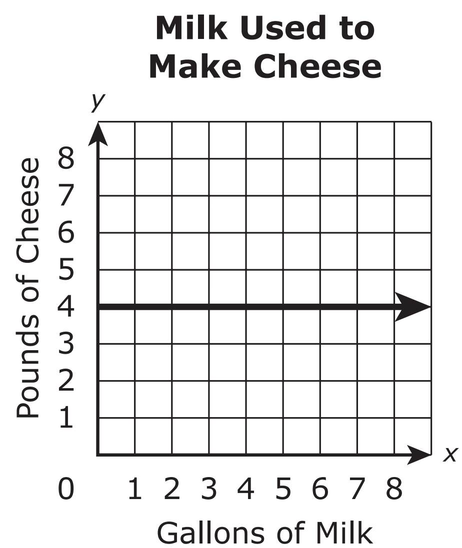
40. A cheese manufacturer uses 65 gallons of milk to make 52 pounds of cheese. The weight of the cheese is directly proportional to the number of gallons of milk used to make the cheese.

Which graph represents the relationship between the number of gallons of milk and the number of pounds of cheese that can be made from the milk?

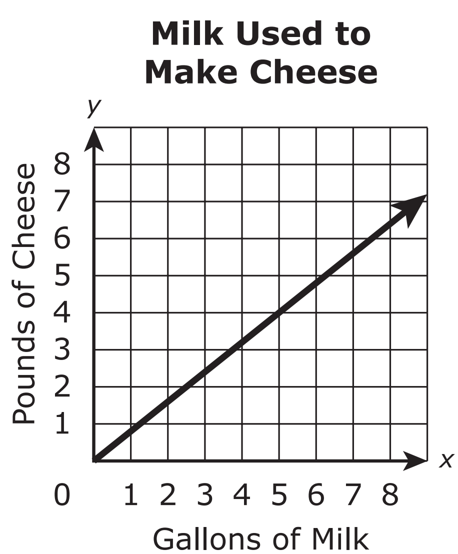
A.



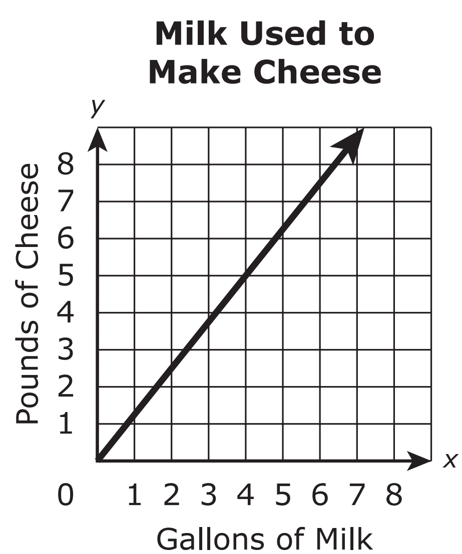
B.



C.



D.



GO ON ►

41. A bakery uses a muffin recipe that uses $\frac{1}{2}$ cup of milk for every batch of 12 muffins.

Part A

Based on the recipe, which statement is true?

Select **each** correct answer.

A. $\frac{1}{24}$ cup of milk is used to make each muffin.

B. $\frac{1}{12}$ cup of milk is used to make each muffin.

C. $\frac{1}{6}$ cup of milk is used to make each muffin.

D. 1 cup of milk is used to make every 6 muffins.

E. 1 cup of milk is used to make every 12 muffins.

F. 1 cup of milk is used to make every 24 muffins.

GO ON ▶

Part B

How many batches of 12 muffins can be made using one **gallon** of milk? Show your work or explain how you found your answer.

Enter your answer and your work or explanation in the box provided.

GO ON ▶

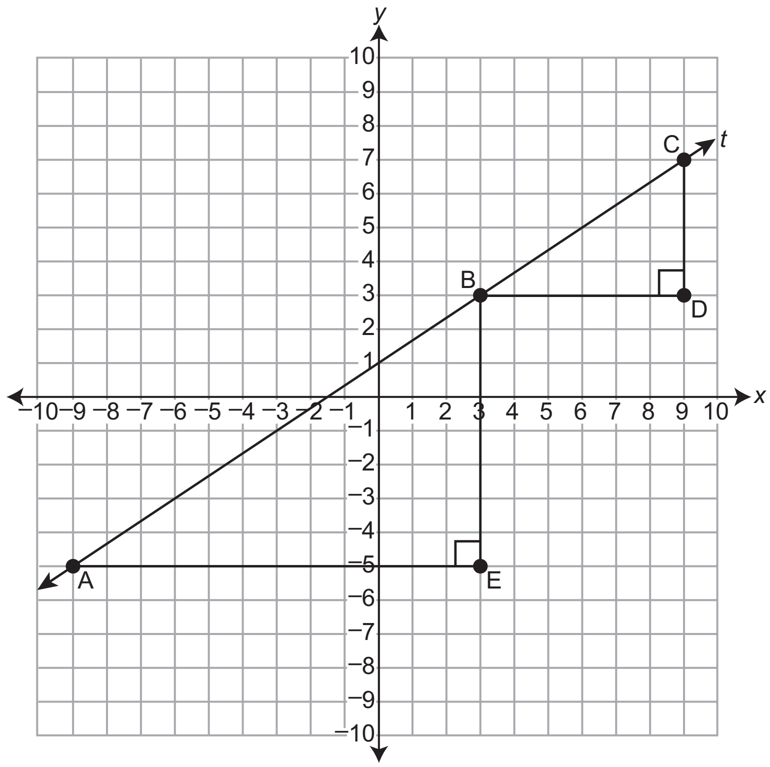
Part C

The bakery makes 96 muffins every day. How many total gallons of milk are needed to make 96 muffins every day for 30 days? Show your work or explain how you found your answer.

Enter your answer and your work or explanation in the box provided.

GO ON ▶

42. Similar triangles *ABE* and *BCD* are shown on the coordinate plane. Line *t* passes through points *A*, *B*, and *C*.



Part A

Select from the list to correctly complete the sentence.

The slope of segment *AB* is ____________________ the slope of segment *BC*.

greater than
less than
equal to

GO ON ►

Part B

Use the ratios of the side lengths of triangle *ABE* and triangle *BCD* to explain your answer to Part A.

Enter your explanation in the box provided.

Part C

Write an equation for line *t*. Show or explain how you determined your equation.

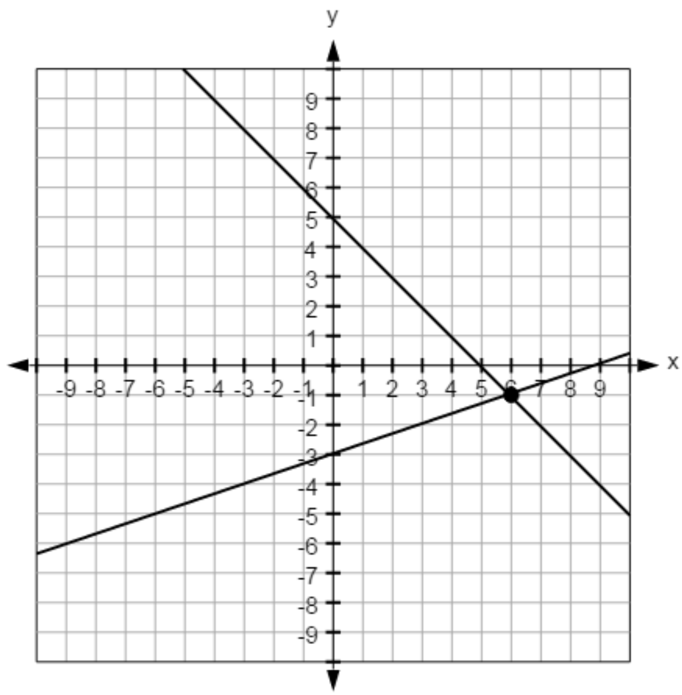
Enter your equation and your work or explanation in the box provided.

STOP

LEAP 2025 Grade 8 Mathematics
Practice Test Answer Key

This document contains the answer keys and rubrics for the LEAP 2025 Grade 8 Mathematics Practice Test.

Session 1				
Task #	**Task Type**	**Value (points)**	**Key**	**Alignment**
1	I	1	B, F	8.EE.A.2
2	I	1	D	8.NS.A.1
3	I	2	Part A: D Part B: C	8.G.A.3
4	I	1	C, D	8.F.A.1
5	I	1	It [does represent a function] because [each input has only one output].	8.F.A.1
6	I	1	D	8.G.A.1b
7	I	2	Part A: A Part B: B	8.G.A.4
8	I	1		8.NS.A.2
9	I	1	A	8.SP.A.2
10	I	1	300	8.EE.A.3

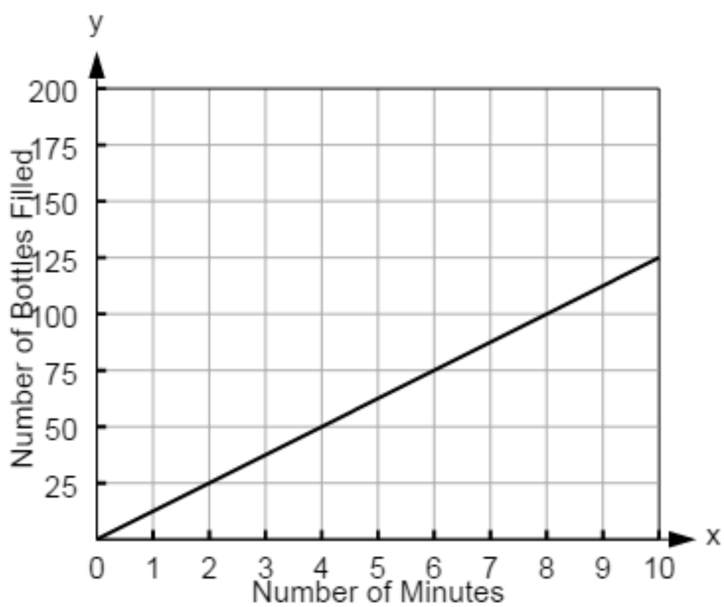
<table>
<tr><th colspan="5">Session 1</th></tr>
<tr><th>Task #</th><th>Task Type</th><th>Value (points)</th><th>Key</th><th>Alignment</th></tr>
<tr><td>11</td><td>I</td><td>1</td><td>
</td><td>8.EE.C.8a</td></tr>
<tr><td>12</td><td>I</td><td>1</td><td>A, C, F</td><td>8.G.A.1a</td></tr>
<tr><td>13</td><td>I</td><td>1</td><td>-1.4</td><td>8.EE.C.7b</td></tr>
<tr><td>14</td><td>I</td><td>1</td><td>10, -2</td><td>8.EE.C.8b</td></tr>
<tr><td>15</td><td>I</td><td>1</td><td><table>
<tr><th></th><th>Increasing</th><th>Decreasing</th><th>Neither Increasing nor Decreasing</th></tr>
<tr><td>$-7 < x < -3$</td><td>✔</td><td></td><td></td></tr>
<tr><td>$-3 < x < -1$</td><td>✔</td><td></td><td></td></tr>
<tr><td>$-1 < x < 1$</td><td></td><td>✔</td><td></td></tr>
<tr><td>$1 < x < 3$</td><td>✔</td><td></td><td></td></tr>
<tr><td>$3 < x < 5$</td><td></td><td>✔</td><td></td></tr>
<tr><td>$5 < x < 7$</td><td></td><td></td><td>✔</td></tr>
</table></td><td>8.F.B.5</td></tr>
<tr><td>16</td><td>I</td><td>1</td><td>D</td><td>8.EE.C.8a</td></tr>
<tr><td>17</td><td>I</td><td>1</td><td>A</td><td>8.EE.A.1</td></tr>
</table>

<table>
<tr><th colspan="5">Session 1</th></tr>
<tr><th>Task #</th><th>Task Type</th><th>Value (points)</th><th>Key</th><th>Alignment</th></tr>
<tr><td>18</td><td>I</td><td>1</td><td><table>
<tr><th></th><th>$y = 7 \times 4x$</th><th>$y = (2x + 5)^2$</th><th>$y = 10x^2$</th><th>$y = 5x - 3$</th><th>$y = \frac{x}{2}$</th><th>$y = 2x^3 + 1$</th></tr>
<tr><td>linear</td><td>✓</td><td></td><td></td><td>✓</td><td>✓</td><td></td></tr>
<tr><td>nonlinear</td><td></td><td>✓</td><td>✓</td><td></td><td></td><td>✓</td></tr>
</table></td><td>8.F.A.3</td></tr>
<tr><td>19</td><td>I</td><td>1</td><td>3.9</td><td>8.EE.A.4</td></tr>
<tr><td>20</td><td>I</td><td>1</td><td>4</td><td>8.EE.C.7b</td></tr>
</table>

Session 2				
Task #	Task Type	Value (points)	Key	Alignment
21	I	1	14.764 or 14.765	8.G.B.7
22	I	1	12	8.G.B.7
23	I	1	A	8.F.A.2
24	I	2	Part A: A Part B: B	8.F.B.4
25	I	1	A	8.EE.B.5
26	I	1	B, C	8.SP.A.4
27	III	3	rubric	LEAP.III.8.3 (8.EE.B.5)
28	I	1	C	8.EE.A.4
29	I	1	The rate of change in Proportion A is 2.5 less than the rate of change in Proportion B.	8.EE.B.5
30	II	3	rubric	LEAP.II.8.2 (8.EE.C.7a, 8.EE.C.7b)
31	II	4	Part A: C Part B: rubric Part C: rubric	LEAP.II.8.3 (7.EE.A.1)
32	III	3	rubric	LEAP.III.8.1 (8.F.A.2, 8.EE.B.5)

Session 3				
Task #	Task Type	Value (points)	Key	Alignment
33	I	1	A, B, C, E	8.EE.B.6
34	I	1	C	8.SP.A.4

Session 3

Task #	Task Type	Value (points)	Key	Alignment
35	I	2	Part A: D Part B: 4.5	8.EE.C.7b
36	I	1	Function B — Least Rate; Function A; Function C — Greatest Rate	8.F.A.2
37	II	3	Part A: rubric Part B: rubric	LEAP.II.8.3 (8.G.A.5)
38	I	1		8.EE.B.5
39	I	2	Part A: C, E Part B: 4.8	8.G.C.9
40	I	1	C	8.EE.B.5
41	III	6	Part A: A, F Part B: rubric Part C: rubric	LEAP.III.8.2 (7.RP.A.1, 7.RP.A.2b, 7.RP.A.3)
42	II	4	Part A: The slope of segment *AB* is [equal to ▾] the slope of segment *BC*. Part B: rubric Part C: rubric	LEAP.II.8.5 (8.EE.B.6)

RUBRICS

<table>
<tr><th colspan="2">Task #27</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>3</td><td>Student response includes the following 3 elements:

Computation component: 2 points

Approximate miles per gallon for car M, from 25 to 27
Approximate miles per gallon for car P, from 28 to 33

Modeling component: 1 point

Valid work shown or explanation given for each answer

Sample Student Response:
Car M gets approximately 26.5 miles per gallon.
I found this by finding an average unit rate for the table for Car M.
50.4 + 80.5 + 181.3 + 137.5 = 449.7 Total Miles
2 + 3 + 7 + 5 = 17 Total Gallons
$\frac{449.7}{17} \approx$ 26.5 Miles Per Gallon

Car P gets approximately 31.7 miles per gallon.
I found this by approximating the points in the graph as
(1, 30), (2, 65), (3, 90), (4, 130) and (5, 160). Then I found the average unit rate for these points.
30 + 65 + 90 + 130 + 160 = 475 Total Miles
1 + 2 + 3 + 4 + 5 = 15 Total Gallons
$\frac{475}{15} \approx$ 31.7 Miles Per Gallon</td></tr>
<tr><td>2</td><td>Student response includes 2 of the 3 elements.</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 3 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
</table>

Task #30	
Score	**Description**

3	Student response includes the following 3 elements:
	• **Computation component**: 1 point
	  ○ Correct explanation of why the conclusion is no solution
	• **Reasoning component**: 2 points
	  ○ Correctly uses the distributive property
	  ○ Correctly combines like terms
	
	Sample Student Response:
	-2(11 – 12x) = -4(1 – 6x)
	-22 + 24x = -4 + 24x
	Subtracting 24x from each side
	-22 + 24x – 24x = -4 + 24x – 24x
	-22 = -4
	This is impossible, since -22 is not equal to -4. Therefore, there is no solution to the equation.

2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

Task #31	
Part B	
Score	**Description**

2	Student response includes the following 2 elements:
	• **Computation component:** 1 point
	  ○ Writes equivalent expressions
	• **Reasoning component**: 1 point
	  ○ Provides a correct series of reasoning to determine that the first expression is always greater than the second expression
	
	Sample Student Response:
	I need to compare the expressions, so I will rewrite them by distributing and combining like terms.
	
	$\frac{1}{2}(7x+48)=\frac{7}{2}x+24$
	and
	$-\left(\frac{1}{2}x-3\right)+4(x+5)=-\frac{1}{2}x+3+4x+20=\frac{7}{2}x+23$
	
	When I compare $\frac{7}{2}x+24$ to $\frac{7}{2}x+23$, I can subtract $\frac{7}{2}x$ from both expressions since they give the same value and just compare 24 to 23. Since 24 is always greater than 23, the expression $\frac{1}{2}(7x+48)$ is always greater than the expression $-\left(\frac{1}{2}x-3\right)+4(x+5)$.
	
	Notes:
	• The student does not need to show both equivalent expressions, but can earn this point if it is clear from their explanation that they found equivalent expressions. For example, if the student explains that the only difference between the two expressions is that one has 23 and the other has 24, it is clear that they have found equivalent expressions.
	• The student may receive a total of 1 point if he or she computes the correct answer, but shows no work or insufficient work to indicate a correct reasoning process.

1	Student response includes 1 of the 2 elements.
0	Student response is incorrect or irrelevant.

Task #31	
Part C	
Score	**Description**
1	Student response includes the following element: • **Modeling component**: 1 point ○ Student creates any expression using the variable *x* that is always greater than the two given expressions. Sample student response: $\frac{7}{2}x + 25$
0	Student response is incorrect or irrelevant.

Task #32	
Score	**Description**

3	Student response includes the following 3 elements:
	• **Computation component**: 1 point
	    ○ Correct unit prices for both gas stations, 4 and 3.80
	• **Modeling component**: 2 points
	    ○ Determines that gas station P charges more for gasoline
	    ○ Correctly models determining the unit prices and the gas station that charges more for gasoline.
	
	Sample Student Response:
	Based on the unit prices, Gas Station P charges more for gasoline. The unit price for Gas Station P is $4.00 per gallon since the constant linear graph for Gas Station P shows the point (5, 20), which means it costs $20 for 5 gallons of gas. The table for Gas Station M shows that 10 gallons cost $38, so the unit price for Gas Station M is $\frac{38}{10}$ = $3.80 per gallon.

2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

Task #37	
Part A	
Score	**Description**
1	Student response includes the following element: • **Reasoning component**: 1 point ○ Correctly reasons why ∠*KJN* and ∠*LJM* are congruent Sample Student Response: ∠*KJN* is congruent to ∠*LJM* because they are the same angle since they exactly overlap.
0	Student response is incorrect or irrelevant.
Part B	
Score	**Description**

2	Student response includes the following 2 elements:
	• **Reasoning component**: 2 points
	  ○ Correct pair of corresponding congruent angles, ∠*JKN* and ∠*JLM* or ∠*JNK* and ∠*JML*
	  ○ Correctly reasons why the given pair of angles is congruent
	
	Sample Student Response:
	∠*JKN* and ∠*JLM*
	OR
	∠*JNK* and ∠*JML*
	
	Either line segment *JK* or line segment *MN* is a transversal to the parallel line segments *KN* and *LM*. When two parallel lines are intersected by a transversal, corresponding angles formed by the transversal are congruent. The pair of angles is also corresponding in terms of their locations in Δ*KJN* and Δ*LJM*.

1	Student response includes 1 of the 2 elements.
0	Student response is incorrect or irrelevant.

<table>
<tr><th colspan="2">Task #41</th></tr>
<tr><th colspan="2">Part B</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>3</td><td>Student response includes the following 3 elements:
Computation component: 1 point
Correct answer, 32
Modeling component: 2 points
Correct strategy to find the total number of cups in a gallon
Correct strategy to find the number of batches of muffins
Sample Student Response:
There are 2 cups in a pint, 2 pints in a quart, and 4 quarts in a gallon, so there are $2 \times 2 \times 4 = 16$ cups in a gallon.

One cup of milk is needed for 24 muffins, so 1 gallon of milk can make
$24 \times 16 = 384$ muffins. This means that $384 \div 12 = 32$ batches of muffins can be made using 1 gallon of milk.

Notes:
Providing the correct number of cups in a gallon is sufficient for modeling component 1.
The student may show modeling using only equations. If the equations shown represent a valid modeling process, credit should be awarded.</td></tr>
<tr><td>2</td><td>Student response includes 2 of the 3 elements.</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 3 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
</table>

<table>
<tr><th colspan="2">Task #41</th></tr>
<tr><th colspan="2">Part C</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>2</td><td>Student response includes the following 2 elements:

Computation component: 1 point
Correct answer, 7.5
Modeling component: 1 point
Correct strategy to find the number of gallons of milk

Sample Student Response:
The bakery makes 96 ÷ 12 = 8 batches of muffins each day. In 30 days, the bakery makes 30 × 8 = 240 batches. Since 32 batches can be made with 1 gallon of milk, 240 batches can be made with 240 ÷ 32 = 7.5 gallons of milk.

Notes:

The student may receive modeling points if the student shows a sufficient modeling process for some or all of the parts indicated but makes one or more computational errors resulting in incorrect answer(s).
The student may receive computation points if he or she computes the correct answer(s) to one or all of the parts but shows no work or insufficient work to indicate a correct modeling process.
The student may not receive more than 2 total points (from parts B and C) for modeling if the explanations, while sufficient to indicate that the student has a correct process, contain nonsense statements.
</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 2 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
</table>

<table>
<tr><th colspan="2">Task #42</th></tr>
<tr><th colspan="2">Part B</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>1</td><td>Student response includes the following element:
• Reasoning component: 1 point
    ○ Correct reasoning using ratios of side lengths

Sample Student Response:
The ratio $\frac{BE}{EA} = \frac{8}{12} = \frac{2}{3}$. The ratio $\frac{CD}{DB} = \frac{4}{6} = \frac{2}{3}$. Since the ratio of the sides of each triangle is $\frac{2}{3}$, the ratios are equal, so $\frac{BE}{EA} = \frac{CD}{DB}$. This means that both segments have the same slope.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
<tr><th colspan="2">Part C</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>2</td><td>Student response includes the following 2 elements:
• Computation component: 1 point
    ○ Correct equation for line t, $y = \frac{2}{3}x + 1$ (or equivalent)
• Reasoning component: 1 point
    ○ Shows or explains that line t has a slope of $\frac{2}{3}$ and a y-intercept of 1

Sample Student Response:
To find the slope of t, I can take any two points on the line and find the ratio of the rise to the run. Using points A and B, I found the slope to be $\frac{3-(-5)}{3-(-9)} = \frac{8}{12} = \frac{2}{3}$. Then I identified the y-intercept of line t by looking at its graph. The line crosses the y-axis at $y = 1$, so the y-intercept is 1.
Therefore, the equation of line t is $y = \frac{2}{3}x + 1$.

Notes:
• The student may receive a combined total of 2 points if the reasoning processes are correct but the student makes one or more computational errors resulting in incorrect answers.
• The student may receive a total of 2 points if he or she computes the correct answers but shows no explanation or insufficient explanation to indicate a correct reasoning.
• The student cannot receive more than 1 point for reasoning (from parts B and C) if the explanations, while sufficient to indicate that the student had correct reasoning, contain nonsense statements.</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 2 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
</table>